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Comparison between interventional versus medical therapy in patients with rheumatic mitral valve stenosis in Tanzania

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Abstract

Background: Rheumatic heart disease remains the most common cardiovascular disease in children and young adults. The outcome of interventional versus medical therapy on the long term is not fully elucidated yet. This study provides contemporary data on the clinical profile, treatment and follow up of patients with rheumatic mitral stenosis (MS) in Tanzania.

Methods: Patients' medical information, investigations and treatment data were recorded in this prospective cohort study. They were followed up for 6–24 months to determine the long-term outcome. Interventional therapy was defined as a combination of surgery and percutaneous balloon mitral valvuloplasty. Kaplan–Meier curves and Cox proportional hazards model were used in analyses. *p*-Value < 0.05 was considered statistically significant.

Results: We enrolled 290 consecutive patients. Interventions were done in half of the patients. Median follow up was 23.5 months. Mortality was higher in the medical than interventional treatment (10.4% vs. 4%, log-rank $p = 0.001$). Median age was 36 years, females (68.3%) and low income (55.5%). Multivalvular disease was found in 116 (40%) patients, atrial fibrillation (31.4%), stroke/transient ischaemic attack (18.9%) and heart failure class III–IV (44.1%). Median (IQR) duration of disease was 3 (4) years, secondary prophylaxis (27.7%) and oral anticoagulants use (62.3%). In multivariable analysis, the risk of death among patients on medical was 3.07 times higher than those on interventional treatment (crude HR 3.07, 95% CI 1.43–6.56, $p = 0.004$), 2.44 times higher among patients with arrhythmias versus without arrhythmias (crude HR 2.44, 95% CI 1.19–4.49, $p = 0.015$) and 2.13 times higher among patients with multivalvular than single valve disease (crude HR 2.13, 95% CI 1.09–4.16, $p = 0.026$).

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Conclusions: Intervention is carrying low mortality compared to medical treatment. Arrhythmias and multivalvular disease are associated with a high mortality. Rheumatic MS is more prevalent in young people, females and individuals with low income. There is a late hospital presentation and a low use of both secondary prophylactic antibiotics and anticoagulants.

KEYWORDS

medical treatment, mortality, Tanzania, predictors, rheumatic mitral stenosis, surgery

1 | INTRODUCTION

1.1 | Global epidemiology

Rheumatic heart disease (RHD) is among the common non-communicable diseases (NCDs) in developing countries. It is estimated to be responsible for up to 1.4 million deaths per year and over 10 million disability-adjusted life-years globally.[1,2] The World Heart Federation (WHF) since 2013 has prioritized RHD as an important public health problem and proposed strategies for its reduction and eradication.[3] However, the proposed strategies need an understanding of the contemporary characteristics of disease presentation and the use of evidence-based interventions in patients residing in endemic areas of RHD.[3]

1.2 | Situation in Tanzania

Despite of high prevalence of RHD in Tanzania,[4] the country was not involved in the two large multicentre RHD registries that have been conducted in the last two decades which covered most of the African countries.[1,5] Registry-based control strategies have been reported to be the best framework for delivering secondary prophylaxis and for the establishment of a countrywide RHD control program. In Tanzania, RHD was not part of national health (research) agendas or strategies despite its intersection with priorities like child and adolescent health, maternal health and NCDs. However, the Ministry of Health has recently initiated special programmes to deal with NCDs including RHD mainly aiming at research, training and treatment.[6]

Mitral stenosis (MS), one of the common forms of RHD, has a peculiar presentation compared with other sub-types of the disease, that is, mitral and aortic regurgitation.[7] Patients present with complications such as heart failure, atrial fibrillation (AF), thromboembolic events and infective endocarditis.[1,8] Owing inadequacy of interventional services, medical treatment is what is commonly initially offered mainly targeting the consequences of the disease such as heart failure and AF.[9] Secondary prophylaxis of ARF and RHD is recommended with a monthly intramuscular benzathine penicillin G (BPG) injection.

Prospective RHD studies have shown that univariate predictors of mortality are heart failure, New York Heart Association (NYHA) functional class and prescription for penicillin[1,8] These studies also showed that males had a higher risk of mortality than females. The current study was conducted to provide contemporary data on the clinical profile, treatment (interventional vs. medical) and follow up of patients with rheumatic MS attended at Jakaya Kikwete Cardiac Institute (JKCI) in Tanzania.

2 | MATERIALS AND METHODS

2.1 | Study design and setting

This was a prospective cohort, single-centre, hospital-based study of Tanzanian patients with rheumatic MS that were attended at the JKCI from August 2019 to May 2022. JKCI is the national/tertiary referral hospital for cardiac diseases and receives patients from all over Tanzania and from different sub-Saharan Africa countries. It has a bed capacity of 104 and offers all services of cardiac disease ranging from congenital to acquired heart diseases including rheumatic heart disease. JKCI offers both invasive and non-invasive cardiac services and on average attending 800 outpatients and 100 inpatients per week. For easier coordination of health services, Tanzania is divided into eight health zones which includes eastern, northern, southern, central, western, lake, southern highlands and Zanzibar. JKCI belongs to the eastern zone and it is currently the only hospital in the country that offer rheumatic heart disease interventional services.

2.2 | Inclusion and exclusion criteria

Symptomatic patients with mitral stenosis were enrolled consecutively from outpatient clinics and inpatients until the sample size was attained. We excluded patients with pure MR, no MS, prosthetic valves and those with other

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forms of non-rheumatic valvular heart disease or other cardiac diseases.

2.3 | Sample size estimation

Two methods for sample size calculation were used: (a) prevalence formula for answering two objectives: (i) to describe the clinical characteristics of patients with rheumatic MS and (ii) to determine the proportion of patients on PBMV, surgery and medical treatment at JKCI. A study done by Makubi et al.[10] among patients with heart failure, found the prevalence of rheumatic MS to be 15%. So, the sample size was calculated from the formula: $n = z^2 p\,(1-p)/\varepsilon^2$, where, n = minimum sample size required, P = prevalence of MS among RHD patients (estimated at 15% according to a study quoted above), ε = margin of error; set at 5% (0.05) and Z = z-score (1.96 at 95% confidence level). The estimated sample size was 196. Adjusting (10%) for non-response estimated sample size was ≈216 patients. and (b) the proportional formula for the objective: to assess the effects of baseline clinical characteristics, cardiac morphology and treatment modality (PBMV, surgery or medical treatment) on patients' outcomes. A study conducted by Cohen and co-workers[11] found that the proportion of patients who underwent operative procedures compared to those who underwent PBMV was 60/64 (93.7%). The sample size was calculated from the formula: $n = z^2 p\,(1-p)/\varepsilon^2$. The estimated sample size was 90. Adjusting (20%) for loss to follow up and non-response, the estimated sample size is ≈110 patients.

2.4 | Data collection

The patient's history was obtained. This included age (6 years and above), residence, sex, education level, occupation and monthly income, date of diagnosis, presenting symptoms, associated comorbidities, medications and interventional history. Physical examination including anthropometric measurements and systemic examination was performed. All patients were clinically evaluated for evidence of rheumatic MS according to recognized clinical and echocardiographic criteria.[12] The severity of mitral stenosis was graded as mild (valve area > 1.5 cm^2), moderate (valve area 1.0–15cm^2) and severe (valve area < 1.0 cm^2). Several echocardiographic (SC 2000 Siemens Echo machine), electrocardiographic (General electronic Mac 400) and roentgenogram data were collected. The echocardiogram examination followed the American Society of Echocardiography (ASE) guidelines.[13] Two-dimensional and M-mode images were used to obtain chamber dimensions and ejection fraction quantification. A venous blood sample was collected for haematological, serological and biochemistry analyses.

2.5 | Treatment strategy

The treatment of patients recruited in this study was following the usual standard of care as per the institute protocol. Patients could either belong to the PBMV, surgery or medical treatment. The definite treatment strategy for patients with severe MS, either PBMV or surgery, was determined during the pre-surgery conference attended by the heart team and was based on echocardiographic data, patient age and comorbidities. Reasons for being on medical treatment included: financial, contraindication to surgery, improved/opted on medications, refusal, waiting for interventions. Details about PBMV and surgical treatment strategies can be found in our previous publications.[14,15] Briefly, the institute adhere to the American College of Cardiology (ACC) and European Society of Cardiology (ESC) guidelines for the management of patients with valvular heart disease.[12,16]

2.6 | Follow up

Patients were followed up at different time intervals for a minimum of 6 months to determine the primary and secondary outcome measures namely mortality and different complications (stroke, infective endocarditis, atrial fibrillation, bleeding, re-intervention) respectively, at the end of follow up. The data were obtained either during patient visits to the outpatient clinic or during hospitalization or by telephone interviews. We defined loss to follow up as a failure to show up/not being able to be contacted 6 months after enrolment and only seen on the initial visit.

2.7 | Statistical analysis

The data obtained from collection tools were entered into the computer and analysed using Statistical Package for Social Sciences (IBM SPSS Statistics) version 28.0 and STATA (StataCorp) version 13. Categorical variables are summarized as numbers and percentages and were compared by chi-squared or Fisher's exact test. Mean (standard deviation) and median (interquartile range) were employed for continuous variables and were compared by Student's t-test. Kaplan–Meier analysis was used to ascertain the incidence of survival probability at the end of follow up. Because of the low number of patients undergoing PBMV, it has been combined with surgery to form the interventional group. Differences between curves were

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described using log-rank tests. Risk factors (hazard ratios with 95% confidence intervals) associated with different outcomes were determined by using the stepwise method of the Cox proportional hazards model. Covariates with $p < 0.2$ in the univariate analysis were added to a multivariable model. Model goodness of fit was assessed using likelihood ratio tests (LRTs). For laboratory variables (not done to all patients) we calculated the incidence rates of mortality as the number of events per 1000 patient-years of follow up. p-Value < 0.05 was considered statistically significant.

2.8 | Ethical consideration

Written informed consent was obtained from all participants ≥18 years. Assent was obtained from minors >13 years of age in the presence of the adult witness. For <13 years, oral consent was provided by the guardian of the minor. The study was approved by the Directorate of Research and Publications of the Muhimbili University of Health and Allied Sciences (P. MUHAS – REC-9-2019-059). Permission to conduct this study was obtained from JKCI (AB.157/334/01'A).

3 | RESULTS

Two hundred and ninety (290) patients were consecutively enrolled in the Tanzania Mitral Stenosis (TAMS) study. Interventional (surgical and PBMV) were done in a half of the patients (Figure 1).

Their baseline sociodemographic and clinical characteristics are shown in Table 1. The two treatment groups were similar except for the level of education, income, ascites, being on beta blockers and benzathine penicillin. There were more females than males and the median (IQR) age was 36[27] years. More than half of the patients attained primary education (62.4%) with many of them in the medical than interventional treatment. There was a monthly income of <42$ in 161 (55.5%) of which many of them were on medical than interventional treatment. Patients owning national health insurance were 131 (45.2%) and 157 (54.1%) were employed. The median (IQR) duration of the disease before presenting to hospital was 36 (48) months. A history of ARF was reported by one-fifth of the patients. The most common symptoms were dyspnoea (97.2%), palpitations (90.3%) and orthopnoea (80.7%). Cardiomegaly was found in 215 (74.1%) patients. At the time of presentation, 128 (44.1%) patients were in heart failure NYHA functional class III–IV, 55 (18.9%) had a previous stroke/transient ischaemic attack, 26 (8.9%) had infective endocarditis, 56 (19.3%) were hypertensive, 91 (31.4%) had atrial fibrillation and 5 (2.5%) were pregnant. A majority, 185 (63.8%) seek medical attention at a modern health facility when they initially experienced symptoms. The commonly prescribed medicines were loop diuretics (85.2%), spironolactone (73.1%) and beta blockers (65.9%). Benzathine penicillin prophylaxis was prescribed in 46 (27.7%) of eligible patients, many belonging the medical treatment. Among patients with AF, 57 (62.3%) were on oral anticoagulants. On the other hand, 25 (45.5%) patients with stroke/TIA and all patients who undergone valve replacement were on oral anticoagulants. Half of the patients were on medical treatment. Reasons for being on medical treatment included: financial issues,[3] contraindication to surgery due to comorbidities/frailty,[13] improved/opted on medications (51), refusal,[18] waiting for surgery/PBMV (60).

As per valvular lesions, 116 (40%) of the patients had multivalvular disease. Mixed mitral valve/single valve disease was present in 174 (60%) patients. Twenty-seven

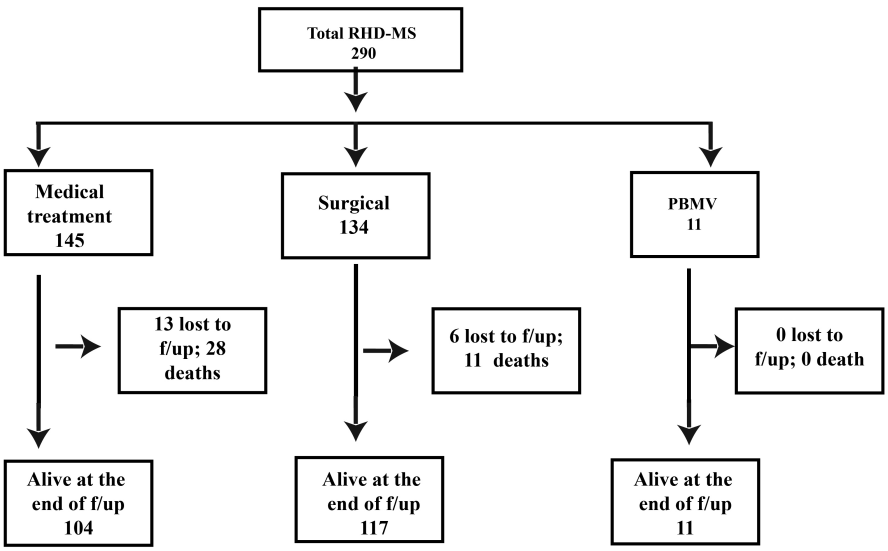


FIGURE 1 A flow chart diagram showing patients recruitment. F/up, follow up; PBMV, percutaneous balloon mitral valvuloplasty; RHD – MS, rheumatic heart disease – mitral stenosis.

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TABLE 1 Socio-demographic and clinical characteristics of the patients with rheumatic mitral stenosis, $N = 290$.

Variable	Category	Total (%)	Intervention (%)	Medical (%)
Age group (years)	<20	50 (17.2)	17 (11.7)	33 (22.8)
	20–40	121 (41.7)	74 (51.0)	47 (32.4)
	≥40	119 (41.0)	54 (37.2)	65 (44.8)
Median age (IQR)		36 (27)	36 (25)	35 (31)
Sex	Male	92 (31.7)	48 (33.1)	44 (30.3)
	Female	198 (68.3)	97 (66.9)	101 (69.7)
Pregnancy		5 (2.5)	0 (0)	5 (3.4)
Level of education	Primary	181 (62.4)	81 (55.9)	100 (69.0)
	Secondary/college	109 (37.6)	64 (44.1)	45 (31.0)
Occupation	Employed	157 (54.1)	82 (56.6)	75 (51.7)
	Not employed	133 (45.9)	63 (43.4)	70 (48.3)
Mode of payment	National insurance	131 (45.2)	69 (47.6)	62 (42.8)
	Public/others	159 (54.8)	76 (52.4)	83 (57.2)
Monthly income (USD)	Low (<42$)	161 (55.5)	69 (47.6)	92 (63.4)
	High (42$+)	129 (44.5)	76 (52.4)	53 (36.6)
Duration of symptoms (months)	<60	192 (66.2)	91 (62.8)	101 (69.7)
	60+	98 (33.8)	54 (37.2)	44 (30.3)
Past history of acute rheumatic fever		53 (18.3)	23 (15.9)	30 (20.7)
Symptoms and signs at enrolment	Dyspnoea	282 (97.2)	140 (96.6)	142 (97.9)
	Fatigue	202 (69.7)	97 (66.9)	105 (72.4)
	Ascites	31 (10.7)	10 (6.9)	21 (14.5)
	Cardiomegaly	215 (74.1)	109 (75.2)	106 (73.1)
	Atrial fibrillation	91 (31.4)	52 (35.9)	39 (26.9)
Cormorbidities	Stroke/TIA	36 (12.4)	20 (13.8)	16 (11.0)
	Infective endocarditis	15 (5.2)	9 (6.2)	6 (4.1)
	Hypertension	56 (19.3)	23 (15.9)	33 (22.8)
NYHA functional class	I and II	162 (55.9)	84 (57.9)	78 (53.8)
	III and IV	128 (44.1)	61 (42.1)	67 (46.2)
Attended at a modern health facility when experienced symptoms		185 (63.8)	102 (70.3)	83 (57.2)
Attended traditional medicine when experienced symptoms		79 (27.2)	38 (26.2)	41 (28.3)
Medications	Loop diuretics	247 (85.2)	127 (87.6)	120 (82.8)
	Beta blockers	191 (65.9)	108 (74.5)	83 (57.2)
	ACEIs	120 (41.4)	61 (42.1)	59 (40.7)
	Benzathine penicillin for prophylaxis	23 (7.9)	6 (4.1)	17 (11.7)

Abbreviations: ACEIs, angiotensin converting enzyme inhibitors; IQR, interquartile range; NYHA, New York Heart Association; TIA, transient ischaemic attack.

(9.3%) patients had pure MS. Secondary triscupid regurgitation was a common presentation in many patients as shown in Figure 2 and in Table 2.

As depicted in Table 2, the two groups were comparable on echocardiographic parameters at baseline. The left ventricular diastolic diameter was normal in the majority, 252 (86.9%) of the patients while the left atrium diameter was dilated in 280 (96.6%) patients. A majority, 234 (80.7%) had normal left ventricular ejection fraction. The majority (72.8%) of the patients had a severe form of mitral stenosis.

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The right ventricular systolic pressure was elevated in 255 (87.9%) patients and the right ventricular function was depressed in 117 (40.3%) patients.

Arrhythmias were detected on ECG in 148 (51%) patients of which 61.5% were due to AF. On chest x-ray, cardiomegaly was reported in 234 (90%) patients and lung congestion in 27 (10.5%) patients. ASOT was elevated in 23 (27.4%) patients, 4 (1.9%) patients had HIV infection, and 8 (5.4%) patients were VDRL positive. Haemoglobin level was low in 44.6% of patients.

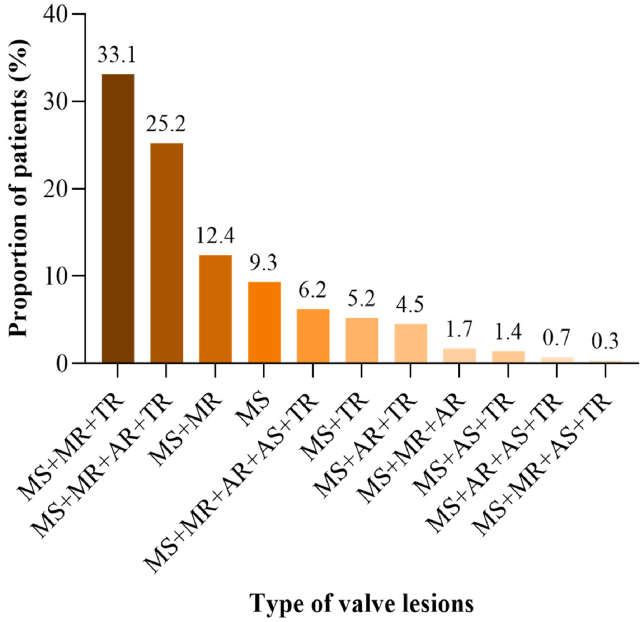


FIGURE 2 Categories of primary pathology of the valves. AR, aortic regurgitation; AS, aortic stenosis; MR, mitral regurgitation; MS, mitral stenosis; TR, tricuspid regurgitation.

3.1 | Follow up

Nineteen (6.6%) patients were lost to follow up. Thus, outcomes data are determined by the remaining 271 patients. These were seen at least for more than one visit with a minimum of 6 months and a maximum of 24 months, that is, varying duration of follow up. The median (interquartile range) duration of follow up was 23.5[26] months.

3.2 | Morbidity

Complications which occurred at any time of follow up were stroke which happened in two (0.7%) patients, 6 (2.2%) patients who developed infective endocarditis and three (1.1%) patients who developed AF. Bleeding due to warfarin toxicity occurred in 10 (3.7%) patients, eight with prosthetic valves and two were not yet operated. One patient who previously underwent PBMV was operated on 2 years later due to progressive symptoms of heart failure.

3.3 | Mortality

There were a total of 39 (14.4%) deaths, 11 (4%) in the interventional treatment arm and 28 (10.4%) in the medical treatment arm. No death occurred in the PBMV group of the interventional treatment arm. As Figure 3 shows,

TABLE 2 Echocardiographic parameters of the patients with rheumatic mitral stenosis, $N = 290$.

Echocardiography parameter	Category	*n* (%)/median (IQR)	Intervention	Medical
LV diastolic diameter (mm)	Abnormal (%)	38 (13.1)	19 (13.1)	19 (13.1)
	Median (IQR)	47 (14)	48.0 (12)	46.0 (15)
Left atrium diameter (mm)	Abnormal (%)	280 (96.6)	142 (97.9)	138 (95.2)
	Median (IQR)	53.9 (12)	54.0 (12)	53.8 (13)
LV ejection fraction (%)	Abnormal (%)	56 (19.3)	30 (20.7)	26 (17.9)
	Median (IQR)	60.4 (12)	60.0 (12)	61.0 (12)
Mitral valve area (cm^2)	Abnormal (%)	211 (72.8)	110 (75.9)	101 (69.7)
	Median (IQR)	0.97 (1)	0.9 (1)	1.0 (1)
Trans-mitral gradient (mmHg)	Median (IQR)	12.9 (4)	13.2 (7)	12.7 (9)
RV systolic pressure (mmHg)	Abnormal (%)	255 (87.9)	132 (91.0)	123 (84.8)
	Median (IQR)	55 (25)	56.0 (21)	55.0 (28)
TAPSE (mm)	Abnormal (%)	117 (40.3)	61 (42.1)	56 (38.6)
	Median (IQR)	17 (4)	17.0 (4)	17.0 (4)
Valvular pathology	Multivalvular disease (%)	116 (40)	64 (44.1)	52 (35.9)

Abbreviations: LV, left ventricle; RV, right ventricle; TAPSE, tricuspid annular plane systolic excursion.

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there was a statistically significant difference in deaths between the interventional and medical treatment arm (log rank *p*-value = 0.001). The mean (SD) age of death was 40.46 (20.09) years. Mortality between females and males was similar (15.5% vs. 11.9%, 95% CI 0.65–2.73, $p = 0.436$). Most deaths occurred within the first 6 months of presentation.

Table 3 shows the univariate analysis of factors associated with mortality. Mortality was higher among self-employed patients (crude HR 8.66, 95% CI 1.16–64.68, $p = 0.035$) and not employed (crude HR 7.73, 95% CI 1.03–57.76, $p = 0.046$) compared with the employed patients; and among patients with a monthly income of <42$ (crude HR 2.23, 95% CI 1.11–4.49, $p = 0.024$) compared to patients with income of >126$. Note a wide 95% CI due to a small number of deaths in the particular groups. The mortality was high among patients who presented with fatigue (crude HR 2.53, 95% CI 1.06–6.04, $p = 0.037$), ascites (crude HR 2.43, 95% CI 1.12–5.29, $p = 0.025$) and hepatomegaly (crude HR 2.12, 95% CI 1.01–4.46, $p = 0.048$) compared with patients who did not present with those symptoms/signs. There was a higher mortality among patients on angiotensin converting enzyme inhibitors (ACEIs) (crude HR 2.37, 95% CI 1.24–4.52, $p = 0.009$) compared with the ones not on ACEIs. Mortalities were higher among patients with arrhythmias versus without arrhythmias (crude HR 2.04, 95% CI 1.05–3.97, $p = 0.036$) and among patients on medical versus interventional treatment (crude HR 3.15, 95% CI 1.57–6.32, $p = 0.001$).

To control for confounding variables, a multivariable analysis was conducted. The independent predictors of mortality were being on medical treatment and presence of arrhythmias as shown in Table 4. The risk of death among patients on medical treatment was 3.07 times higher than those on interventional treatment (crude HR 3.07, 95% CI 1.43–6.56, $p = 0.004$), 2.44 times higher among patients with arrhythmias vs without arrhythmias (crude HR 2.44, 95% CI 1.19–4.49, $p = 0.015$) and 2.13 times higher among patients with multivalvular than single valve disease (crude HR 2.13, 95% CI 1.09–4.16, $p = 0.026$).

Table 5 shows that the incidence rate of deaths were statistically significant higher among patients with elevated antistreptolysin O titres, abnormal white blood cells, low haemoglobin level, elevated C-reactive protein, elevated creatinine, low sodium and elevated alanine transaminase compared to patients with normal values. Note a wide 95% CI due to small number of deaths in the particular groups.

4 | DISCUSSION

This is the first follow up study on the outcomes of rheumatic mitral stenosis in Tanzania. The study has seven main findings. First, patients were young (median age 36 years), predominantly females (68.3%), and had a monthly income <42$ (55.5%). Second, patients presented with a long duration of disease (median duration 3.8 years) and there was low use of secondary antibiotic prophylaxis (27.7%). Third, interventions were done in half of the patients (46.2% surgical and 3.8% PBMV). Fourth, there were 11 (4%) deaths in the interventional treatment and 28 (10.4%) in the medical treatment group. Fifth, the independent predictors of mortality were being on medical treatment, presence of arrhythmias (predominantly AF) and presence of multivalvular disease.

The current study has shown that rheumatic MS affects young patients (median age of 36 years). Similarly, In REMEDY[1] and VALVAFRIC registries,[5] the median age of the patients was 28 years. The relatively higher median age in our study compared with the two registries is because we recruited patients aged 6–80 years while the registries recruited patients aged 5–65 years. Female predominance is a well-recognized finding in RHD

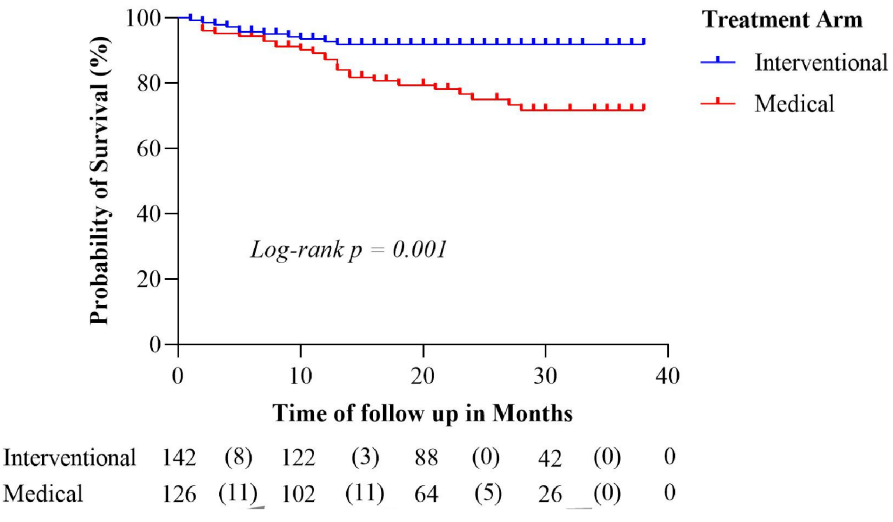


FIGURE 3 Kaplan–Meier curves of participant survival stratified by treatment arm.

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TABLE 3 Univariate analysis of factors associated with mortality among patients with rheumatic MS.

Variable	Category	cHR	95% CI of cHR	*p* Value
Age (years)	≥41	1.14	0.48–2.72	0.763
	20–40	0.75	0.30–1.87	0.530
	<20	Ref		
Sex	Female	1.33	0.65–2.73	0.436
Level of education	No formal to primary	3.04	1.34–6.89	0.008
	Secondary and above	Ref		
Occupation	Not employed	7.73	1.03–57.76	0.046
	Self employed	8.66	1.16–64.68	0.035
	Employed	Ref		
Monthly income (USD)	Low	2.23	1.11–4.49	0.024
	High	Ref		
Duration of illness	≥60 months	0.76	0.38–1.50	0.428
	<60 months	Ref		
Symptoms and signs	Dyspnoea	1.12	0.15–8.14	0.913
	Fatigue	2.53	1.06–6.04	0.037
	Ascites	2.43	1.12–5.29	0.025
	Hepatomegaly	2.12	1.01–4.46	0.048
Medications	Loop diuretics	2.18	0.67–7.08	0.195
	Spironolsctone	2.41	0.94–6.15	0.067
	Beta blockers	0.66	0.35–1.24	0.197
	ACEIs	2.37	1.24–4.52	0.009
	ARBs	0.39	0.09–1.62	0.196
Electrocardiography	Arrhythmias	2.04	1.05–3.97	0.036
Echocardiography	Dilated LV in diastole	1.22	0.51–2.91	0.655
	Dilated left atrium	0.89	0.12–6.56	0.916
	Reduced LV ejection	1.51	0.74–3.09	0.262
	Depressed TAPSE	1.65	0.88–3.09	0.121
Valvular pathology	Multivalvular disease	1.83	0.97–3.43	0.060
	Single valve disease	Ref		
NYHA function classification	III and IV	1.88	0.99–3.55	0.053
	I and II	Ref		
Treatment arm	Medical	3.15	1.57–6.32	0.001

Abbreviations: ACEIs, angiotensin converting enzyme inhibitors; ARBs, angiotensin receptor blockers; cHR, crude hazard ratio; CI, confidence interval; LV, left ventricle; Ref, reference category; TAPSE, tricuspid annular plane systolic excursion.

studies.[1,5,8,17,18] It is not uncommon for women to be diagnosed with RHD for the first time when they become pregnant. In the current study, five women were pregnant when they were diagnosed with RHD for the first time. In RHD, the risk of pregnancy is increased – management of a pregnant patient with RHD is complicated and there is an associated increase in maternal death.[1] We have recently published from our RHD hospital cohort that, the majority of women of reproductive age are at the highest pregnancy risk based on the modified WHO classification and few of them are on contraception.[19] This calls for strategies aiming at reducing these risks that include the provision of family planning and pre-pregnancy advice for women with RHD.

Our study showed that patients had a long duration of disease before presenting to the hospital indicating that

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TABLE 4 Multivariable analysis of the factors associated with mortality among patients with rheumatic MS.

Variable	Category	aHR	95% CI of aHR	*p*-Value
Monthly income (USD)	Low	1.11	0.51–2.43	0.794
	High	Ref		
Education	No formal to primary	1.95	0.81–4.66	0.135
	Secondary and college	Ref		
Fatigue	Yes	1.02	0.39–2.70	0.964
	No	Ref		
Ascites	Yes	0.98	0.41–2.36	0.970
	No	Ref		
Treatment arm	Medical	3.07	1.43–6.56	0.004
	Interventional	Ref		
Loop diuretics	Yes	1.53	0.39–6.05	0.546
	No	Ref		
Spironolactone	Yes	2.50	0.89–7.03	0.083
	No	Ref		
Beta blockers	Yes	0.55	0.27–1.13	0.104
	No	Ref		
ACEIs	Yes	1.67	0.81–3.45	0.163
	No	Ref		
ARBs	Yes	0.50	0.11–2.23	0.365
	No	Ref		
TAPSE (mm)	Normal	0.88	0.45–1.73	0.711
	Abnormal	Ref		
Presence of arrhythmias on ECG	Abnormal rhythm	2.44	1.19–4.99	0.015
	Normal sinus rhythm	Ref		
Valvular pathology	Multivalvular disease	2.13	1.09–4.16	0.026
	Single valve disease	Ref		
NYA function classification	III and IV	0.74	0.36–1.54	0.425
	I and II	Ref		

Abbreviations: ACEIs, angiotensin converting enzyme inhibitors; aHR, adjusted hazard ratio; ARBs, angiotensin receptor blockers; CI, confidence interval; ECG, electrocardiography; Ref, reference category; TAPSE, tricuspid annular plane systolic excursion.

they present for care at a very late stage of the disease. At presentation, 97% reported dyspnoea of any severity, AF (31%), NYHA class III–IV (44%), stroke/TIA (19%), infective endocarditis (9%), severe MS (75%) and multivalvular disease in a majority. These results are similar to those reported in previous studies.[1,8,18] Interestingly, 64% of the patients seek medical attention at a health facility when they initially experienced symptoms highlighting that probably the delay is at lower healthcare facility levels. It is not uncommon for patients with symptoms of heart failure to be misdiagnosed as patients with pulmonary disease, especially when attended by inexperienced general practitioner. This informs that there is a need for capacity building by ensuring proper training to healthcare workers for early detection of the disease at lower levels. In the current study, secondary antibiotics prophylaxis was prescribed in a minority (27.7%) of the patients. Some of the reasons for the low uptake were lack of BPG prescription, believing not in need of BPG, painful injection and lack of knowledge about the disease and prevention. In contrast, the intake of prophylaxis in Ugandan and REMEDY registries were 67% and 60%, respectively. Recently, a trial has shown that secondary prophylaxis retards the progression of the disease.[20] Low uptake of secondary prophylaxis has been reported in several LMICs[1] underscoring the need to identify barriers and enhance its access within the framework of RHD control programmes in endemic areas.[21]

In the current study, half of the patients received interventions (surgery and PBMV). Outcomes of patients who have undergone PBMV[15] and surgery[22] in RHD hospital cohort have been recently published. This is a very interesting and encouraging finding given the fact that in many low- to middle-income countries (LMICs) most of these patients are likely to be managed conservatively due to the

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TABLE 5 Mortality rate stratified by laboratory parameters of the patients with rheumatic MS.

Variable		Total (*N*)	Person time	Death (*n*)	Incidence rate per 1000 patients	95% CI of Incidence
ASOT (units/mL)	Normal	61	1685	5	2.97	1.24–7.13
	High	23	549	7	12.75	6.08–26.75
WBCs	Normal	178	4069	19	4.66	2.97–7.31
	Abnormal	54	954	11	11.51	6.37–20.78
Haemoglobin (g/dl)	Low	103	2248	19	8.43	5.38–13.22
	Normal	122	2685	11	4.10	2.27–7.40
CRP (mg/L)	Normal	62	1445	9	6.23	3.24–11.97
	High	124	2721	19	6.98	4.45–10.95
Creatinine (μmol/L)	Normal	211	4630	27	5.83	4.00–8.50
	Elevated	14	240	4	16.67	6.26–44.41
Sodium (mmol/L)	Normal	80	1859	15	8.07	4.86–13.38
	Low	123	2659	13	4.89	2.84–8.42
ALAT (U/L)	Normal	128	3066	17	5.54	3.45–8.92
	Elevated	43	830	6	7.23	3.25–16.09

Abbreviations: ALT, alanine transaminase; ASOT, anti-streptolysin O titre; CR*P*, C-reactive protein; WBC, white blood cells.

inadequacy of these services. It should be noted that even though the remaining half of the patients were on medical treatment, not all of them required interventions due to several reasons, implying that the uptake of interventions in this cohort was satisfactory. Reasons to why half of the patients were on medical treatment include being waiting for interventions, refusal, not meeting criteria for surgery and financial constraints among others. Patients being on waiting list is attributed by the reason (among others) that JKCI is the only centre in the country that is offering interventions for other conditions such as congenital heart diseases, structural heart diseases and other valvular diseases apart from mitral stenosis. The medicines include diuretics, beta-blockers, digoxin (if in AF), ivabradine (if in sinus rhythm), warfarin for patients with significant rheumatic MS with AF and non-vitamin K antagonist oral anticoagulants for patients without significant MS with AF.[9] In REMEDY,[1] and VALVAFRIC[5] registries, although most the patients were likely to require intervention, only 10.3% of them received the interventions. Similarly, in South Africa[23] and Uganda,[24] interventions were provided in 22% and 9% of the patients who required it respectively. Timely provision of these interventions might improve outcomes.[12] There is a need of implementing strategies for providing high-quality care at tertiary-level facilities for RHD patients in Africa.[25]

In our study, the mortality of 14.3% is somewhat similar to the 11% observed in REMEDY but lower than the 18% reported in the Ugandan[8] and 16% in VALVAFRIC[5] cohorts. The high mortality in the current study is mainly contributed to the patients who were on medical treatment, an independent predictor. Similarly, studies have shown that if left untreated the prognosis of RHD is worse—imposing death at an early age with a mean age of death of 26 years.[26] The mean (SD) age of death in our study was 40.46 (20.09) years because we recruited patients ranging from 6 to 80 years. In fact, the 4% mortality seen in patients who undergone surgery is similar to that reported in a study done by Panda et al.[27] Interestingly, a tremendous improvement in surgical outcomes is observed when compared with the 14.1% from the same institution 10 years ago.[18] However, the low mortality in a surgical group may be due to selection bias in which surgeons may opt for lower risk patients. Percutaneous balloon mitral valvuloplasty (PBMV) is a treatment of choice for suitable candidates offering symptomatic relief lasting up to 20 years.[28] Surgery, individualized based on the patient's health status, surgical capabilities, the extent of valve damage, patient preferences, geographic background and socio-cultural and economic background should be performed for patients who are not suitable for PBMV.[9] These findings underscore a need of improving access to cardiac care services for RHD in Africa.[29]

This study showed that arrhythmias (predominantly AF) were an independent predictor of mortality. In MS there is a decreased LV filling, end diastolic volume and LV stroke volume. Arrhythmias further decreases diastolic filling time resulting in worsening of symptoms ± death. In our study, the mortality was 2.44 times higher in patients with arrhythmias compared with patients without arrhythmias. Similarly, in REMEDY registry, the mortality was 1.4 times higher in patients with AF compared with patients without AF.[1] Several other studies conducted in sub-Saharan Africa (SSA)

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had consistently shown that arrhythmias, particularly AF, are a major public health problem and a common cause of morbidity and mortality.[1] In the current study, the prevalence of AF was 31% comparable with those reported from other studies in SSA,[1] but lower compared with the reported 40% from high-income countries.[30] The postulated reasons for the difference in prevalence are patients from developed countries have a higher life expectancy and higher chronicity of heart failure and patients from SSA are young. Recently, we have published from our hospital RHD cohort that arrhythmias were detected on 12-lead ECG in 49.5% of RHD patients highlighting a need for its early detection and management.[31] Indeed, we had previously reported that there is unmet need in management of arrhythmias among heart failure patients in Africa.[32]

In this study, the proportion of patients with AF was 31.4% which is similar to the approximated 30% in some studies[1,33,34] but lower than the 60% that was reported in another study.[35] Among patients with AF, oral anticoagulants were prescribed in 62.3%, which is slightly lower than the 68.6% use of oral anticoagulants that was reported in the REMEDY registry.[1] Factors such as poor awareness of the importance of anticoagulation among both physicians and patients, poor patient education and limited access to international normalized ratio (INR) testing have been reported to contribute to the sub-optimal utilization of anticoagulants.[1] As per our institute protocol, all patients with mechanical valve prosthesis are put on oral anticoagulants with warfarin with follow up of INR targeting therapeutic range. For single mitral valve replacement, target INR is 2–2.5 and for double valve replacement the target INR is 2–3. As vitamin K antagonist oral anticoagulants remain the main stay among RHD patients with indications of anticoagulation, there is a need of ensuring availability of point-of-care INR devices at different levels of health facilities, including primary health care. Currently, in sub-Saharan Africa, there are great challenges with anticoagulation, with wide variation in its use and time in the therapeutic range of 27–56%.[32,36]

The current study showed that multivalvular disease was present in 116 (40%) patients. This number is higher and lower than a 13% and 64.8% reported in VALVAFRIC[5] and REMEDY[1] registries, respectively. The observed difference could be due to the difference in study population. Moreover, in this study multivalvular disease was an independent predictor of mortality. On the contrary, in the REMEDY registry,[1] multivalvular disease was not an independent predictor of mortality. Data on multivalvular disease are few because of a large number of possible combinations and difficulties of exact quantification. As a result, many areas related to multivalvular disease are not covered by the Guidelines on Valvular Heart Disease issued by the ESC[12] or the ACC.[16] The clinical picture is mainly determined by the combined haemodynamic effect resulting from the severity of each single lesion. The lack of data does not allow the proposal of a standardized, evidence-based management approach. A careful quantification considering diagnostic shortfalls and a focused assessment of the valvular lesion are needed for optimal therapeutic approach for every patient.

Lastly, previous studies had found that age, sex, education level, heart failure, New York Heart Association (NYHA) functional class and left ventricular diastolic diameter were independent predictors of mortality.[1,8] None of these factors showed significant association with mortality in our study. It is possible that this could be due to the difference in studied population. While the referenced studies recruited patients with RHD regardless of the nature of valve lesions, we recruited patients with a predominant mitral stenotic lesion. Moreover, the median age of our study population (36 years) differed from the reported in those studies (28 years). In this study, there was no difference in mortality between males and females. Similarly, while some studies have shown that females had a higher risk of mortality than males,[37] others[1,8] reported the opposite. Indeed, in our previous review, we concluded that it is not clear whether sex plays a role in the diagnosis, management and prognosis of MS and hence a need for research to explore the mechanisms for the differences.[17] Our study showed that patients who were self/not employed and those with low income had a higher risk of death compared with employed patients and patients with better income probably due to access to better healthcare through employer. Similarly, studies that correlated socioeconomic status with RHD outcomes had reported lower mortality among civil servants.[1,5,8] Improved socioeconomic status has been recommended in RHD control programmes.[21] Our study have shown that presenting with fatigue, ascites and hepatomegaly were univariate predictors of mortality similar to what have been reported from previous studies indicating initial presentation with advanced disease.[1,5,8] In this study, the incident rates of death were higher among patients with elevated ASOT and elevated CRP compared to patients with normal values. Similarly, in a study done by Okello et al, elevated ASOT and CRP were predictors of mortality.[8]

5 | STRENGTHS AND LIMITATIONS

Our study has several advantages. First, it is a prospective study with a reasonable follow up period for ascertaining outcomes of interest. Second, it provides contemporary baseline data for future comparisons/

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planning. Third, it has an adequate sample size with the power to allow satisfactory statistical findings. Despite the fact that JKCI receives cardiac patients from all regions of the country, it is necessary that our results may not be generalized for the entire country because there may be factors associated with the way patients access the cardiac services that may hinder representation of patients based on probability samples. Moreover, the selection bias is playing a role in the chosen method of treatment.

6 | CONCLUSION AND RECOMMENDATION

Intervention is carrying low mortality compared to medical treatment and arrhythmias and multivalvular diseases are associated with high mortality. In Tanzania, rheumatic MS affects young people, females and with low income. There is late hospital presentation, low secondary prophylaxis utilization and suboptimal oral anticoagulants use. A half of patients were on medical treatment with two-fifth waiting for interventions. We recommend optimization of interventional services, secondary prophylaxis utilization, oral anticoagulants use, managing arrhythmias and consideration of patients with multivalvular disease during management.

AUTHOR CONTRIBUTIONS

RKM, PC and SC conceptualized and planned for the study. RKM, HM, EK, ME, PK, TS and EN took charge of the data collection and data processing. RKM, PPK, AB, SC, AK, PC, GK and MC supported the analysis, write-up and review of the manuscript. All authors approved the final manuscript for submission.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data used in this paper are available on request from the corresponding author.

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